

Requirement Analysis

Data Flow Diagram & User Stories

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Data Flow Diagram

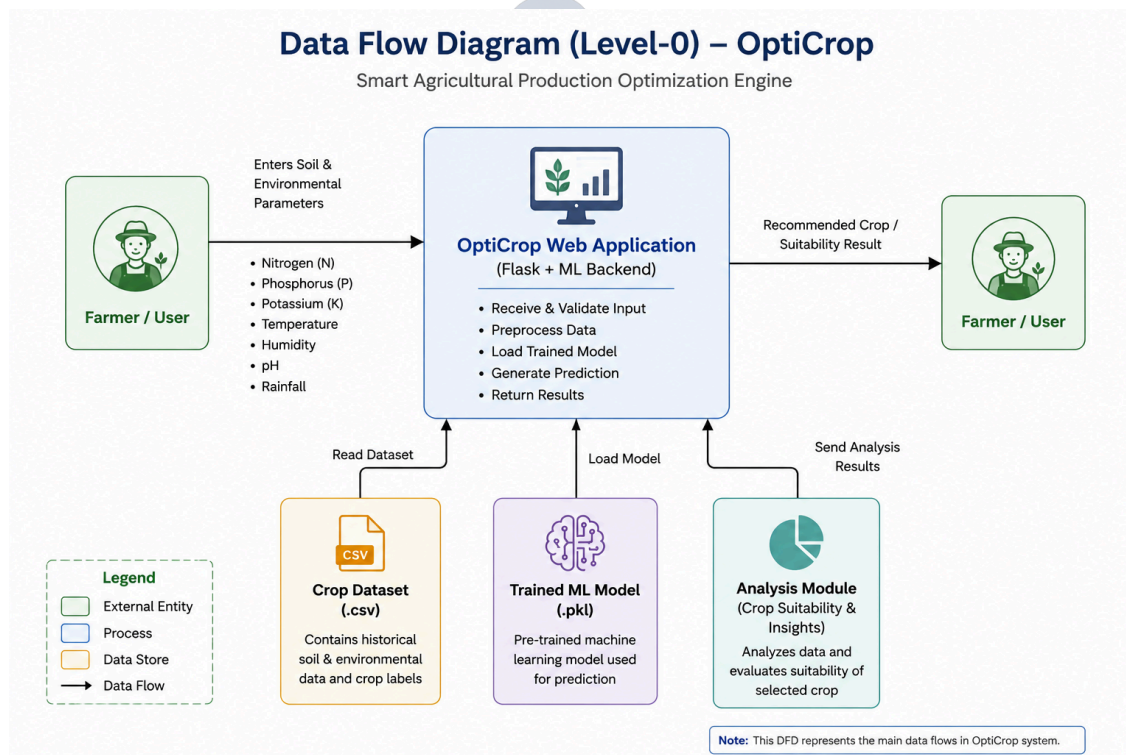


Figure 1: Level-0 Data Flow Diagram of Smart Agricultural Production Optimization Engine (OptiCrop)

User Stories

The following user stories describe the functional requirements of the proposed solution from the perspective of different users.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Farmer	Crop Recommendation	USN -1	As a farmer, I want to enter soil nutrient and environmental parameters so that I can receive the most suitable crop recommendation.	The system accepts valid input and displays the recommended crop.	High	Sprint-1
Farmer	Crop Recommendation	USN -2	As a farmer, I want the system to validate my inputs before prediction.	Invalid or incomplete inputs are identified and appropriate messages are displayed.	High	Sprint-1
Farmer	Crop Suitability Evaluation	USN -3	As a farmer, I want to check whether a specific crop is suitable for my current soil and	The system evaluates the selected crop and displays its suitability.	High	Sprint-2

			weather conditions.			
Agricultural Researcher	Dataset Analysis	USN -4	As an agricultural researcher, I want to analyze crop datasets to understand relationships between soil parameters and crop suitability.	The system provides dataset analysis and model comparison results.	Medium	Sprint-2
System Administrator	Model Management	USN -5	As a system administrator, I want to train and compare multiple machine learning algorithms to select the best-performing model.	The system evaluates algorithms and stores the best model.	High	Sprint-1
System Administrator	Model Deployment	USN -6	As a system administrator, I want to save and load the trained machine learning model for prediction without retraining.	The application successfully loads the saved model (.pkl) and generates predictions.	High	Sprint-1
Farmer	Prediction Results	USN -7	As a farmer, I want to clearly view the recommended crop after submitting my inputs.	The recommended crop is displayed in a simple and understandable format.	High	Sprint-1

